Mansoura University

Faculty of Computers and Information

Department of Computer Science



BUNORA

Coffee Quality Prediction Using Machine Learning

"Every Bean Has a Story."

Project Category: Machine Learning / Full-Stack Application

Course Name: AI / Machine Learning Project

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Submission Date: October 11, 2026

Academic Project Proposal for Machine Learning

Project Team Members

The following research and development team constitutes the BUNORA project group. All members are enrolled in the Faculty of Computers and Information at Mansoura University.

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1 Abstract

Assessing coffee bean quality is traditionally a labor-intensive, subjective, and highly specialized process carried out by certified Q-graders through sensory evaluation. While essential for the global coffee trade, this manual pipeline introduces bottlenecks and variability. This project proposes **BUNORA**, an integrated machine learning application designed to predict coffee quality scores based on verifiable agronomic, geographic, and physical metadata available prior to tasting. Utilizing the Coffee Quality Institute (CQI) dataset, BUNORA establishes a rigorous end-to-end machine learning pipeline encompassing data auditing, automated preprocessing, feature engineering, and empirical comparison of diverse supervised regression models ranging from linear baselines to advanced ensemble algorithms. To bridge the gap between theoretical modeling and practical utility, BUNORA features a robust deployment architecture comprising a Python FastAPI backend and a responsive React TypeScript frontend. This proposal outlines the architectural blueprint, methodological rigor, ethical considerations regarding target leakage, and a structured eight-week implementation timeline.

2 Introduction

Coffee is one of the most widely consumed beverages and traded commodities globally, supporting the livelihoods of millions of smallholder farmers across tropical regions. The economic value of coffee is inextricably linked to its quality, which dictates market pricing, consumer preferences, and brand positioning. Historically, quality assessment relies on expert sensory panels (cuppers) evaluating attributes such as aroma, flavor, acidity, and body.

However, sensory evaluation is inherently subjective and dependent on expert availability. With the growing availability of agricultural, geographic, and processing metadata, data-driven approaches offer a complementary mechanism to anticipate coffee quality before physical tasting occurs. **BUNORA** investigates whether historical coffee metadata can reliably predict cup quality scores, providing an automated tool for producers, exporters, and roasters to evaluate bean batches prior to formal cupping sessions.

3 Background and Project Motivation

The global coffee industry differentiates beans primarily through botanical species (Arabica and Robusta), geographic origin, altitude, processing methods (washed, natural, honey), and physical attributes such as moisture content and defect counts. The Coffee Quality Institute (CQI) maintains rigorous standardized datasets recording these parameters alongside detailed cupping scores [1, 2].

The primary motivation behind BUNORA is twofold:

1. **Agricultural Decision Support:** Helping coffee producers understand which controllable agronomic factors (e.g., altitude, processing technique) most significantly influence cup quality.
2. **Streamlined Supply Chain Operations:** Enabling importers and roasters to quickly screen coffee offerings based on preliminary metadata before investing in physical sample shipments.

4 Problem Statement

Assessing coffee quality involves complex interactions between environmental factors, bean genetics, and post-harvest processing. Machine learning can help identify useful mathematical

relationships between available coffee metadata and historical quality scores.

The proposed system investigates whether coffee metadata can support useful quality-score prediction, compares different algorithms under a consistent evaluation methodology, and identifies which features contribute most to model predictions. It is crucial to distinguish between two distinct prediction tasks:

- **Pre-Tasting Metadata Prediction (Primary Focus):** Predicting the final quality score using only features available prior to tasting (e.g., country of origin, altitude, variety, processing method, moisture, and defect counts).
- **Post-Tasting Sensory Estimation (Optional Extension):** Estimating overall quality when component sensory scores (aroma, flavor, acidity, etc.) are already known.

BUNORA's primary experiment focuses exclusively on pre-tasting metadata to ensure practical predictive utility.

5 Proposed Solution

BUNORA provides a complete, proposed end-to-end machine learning application structured into three integrated tiers:

1. **Machine Learning Core:** A scikit-learn pipeline performing robust data cleaning, missing-value imputation, feature encoding, scaling, model training, and hyperparameter tuning [3].
2. **Backend REST API:** A FastAPI service providing endpoints for model inference, health checks, and metadata retrieval, backed by Pydantic validation [4].
3. **Frontend Web Application:** A responsive React and TypeScript interface styled with Tailwind CSS, enabling users to input coffee characteristics and receive instant, interpretable quality predictions [5].

6 Project Aim and Specific Objectives

The primary aim of BUNORA is to demonstrate the complete machine learning lifecycle applied to agricultural economics. Specific objectives include:

- Conducting a thorough data audit and exploratory data analysis (EDA) on the CQI coffee database.
- Designing an automated preprocessing pipeline that prevents data leakage.
- Training and evaluating multiple regression models, establishing a robust baseline, and performing hyperparameter optimization.
- Implementing model explainability techniques (permutation importance) to identify key quality drivers.
- Developing a FastAPI backend and a React frontend to operationalize the trained model.

7 Research Questions

- **RQ1:** To what extent can pre-tasting agronomic and geographic metadata predict final coffee cup quality scores?
- **RQ2:** Which machine learning model architecture (linear models versus tree-based ensembles) achieves superior generalization performance on mixed-type agricultural tabular data?
- **RQ3:** Which specific environmental or processing features contribute most significantly to predicted coffee quality?

8 Research Scope and Assumptions

The project scope is restricted to publicly available CQI dataset records covering Arabica and Robusta beans. Assumptions include:

- Historical cupping scores in the dataset serve as a consistent proxy for quality.
- Input metadata provided by users via the web interface adheres to valid physical and geographical ranges.
- The project does not aim to replace certified Q-graders, but rather to provide an auxiliary estimation tool.

9 Dataset Description and Data Source

The primary dataset is the Coffee Quality Database associated with the Coffee Quality Institute (CQI), accessible via GitHub and Kaggle [1, 2].

The referenced dataset contains approximately 1,340 coffee reviews, comprising roughly 1,312 Arabica records and 28 Robusta records [1, 2]. These figures represent reported statistics; exact usable observation counts will be confirmed during dataset auditing. Key attributes include species, country of origin, region, altitude, variety, processing method, moisture, defect counts, and component sensory scores.

10 Proposed Machine Learning Methodology

The machine learning workflow follows a strict sequential pipeline: Dataset acquisition → Data audit → EDA → Data cleaning → Missing-value analysis → Duplicate detection → Feature selection → Categorical encoding → Numerical preprocessing → Train/validation/test split → Baseline model → Algorithm comparison → Hyperparameter tuning → Evaluation → Feature importance → Pipeline serialization → API deployment.

Target Variable and Feature Specification

To ensure methodological clarity and prevent target leakage, the machine learning components are explicitly structured as follows:

- **Target Variable:** `total_cup_points`, subject to confirmation of its meaning and distribution during the initial dataset audit [1, 2].
- **Input Features:** Pre-tasting agronomic, geographic, and physical metadata including country of origin, altitude, coffee variety, processing method, moisture content, and defect counts.

- **Excluded Features:** All component sensory scores—such as aroma, flavor, aftertaste, acidity, body, balance, uniformity, clean cup, and sweetness—along with any duplicate or derived versions of the target, ensuring complete avoidance of target leakage.

11 Data Preprocessing and Exploratory Data Analysis

Exploratory Data Analysis will investigate feature distributions, skewness in altitude and moisture measurements, and class imbalance between Arabica and Robusta samples. Preprocessing includes:

- **Missing-Value Imputation:** Median imputation for continuous numerical features (e.g., altitude, moisture) and mode or constant placeholder imputation for categorical attributes.
- **Categorical Encoding:** One-hot encoding for low-cardinality nominal features and target/frequency encoding where appropriate for high-cardinality attributes like region or farm name.
- **Leakage Prevention:** All imputation and scaling parameters will be computed exclusively on training folds using scikit-learn `Pipeline` and `ColumnTransformer` [3].

12 Feature Engineering

Feature engineering will focus on extracting meaningful signals from raw text and numerical ranges:

- Parsing and standardizing altitude ranges into mean altitude values.
- Grouping sparse categorical variables (e.g., rare countries or varieties) into broader regional or botanical categories.
- Calculating total defect metrics by aggregating category one and category two defect counts.

13 Models and Algorithm Comparison

To identify the most effective predictive model, BUNORA evaluates a diverse set of regression algorithms [3]:

1. **DummyRegressor:** Serves as a naive baseline predicting central tendency.
2. **Ridge Regression:** Linear model with L2 regularization to prevent multicollinearity issues among metadata features.
3. **Random Forest Regressor:** Bagging ensemble robust to outliers and non-linear relationships.
4. **Extra Trees Regressor:** Extremely randomized trees reducing variance further.
5. **Gradient Boosting Regressor:** Sequential boosting model capturing complex interactions.
6. **XGBoost / CatBoost:** Advanced gradient boosting frameworks optimized for tabular data structures.

14 Training, Validation, and Testing Strategy

The dataset will be partitioned using an 80/20 train-test split. The training partition will undergo k-fold cross-validation ($k = 5$) for hyperparameter tuning and model selection. Evaluation will be conducted on the held-out test set to ensure unbiased generalization estimates.

15 Evaluation Metrics

Model performance will be assessed using three standard regression metrics [3]:

- **Mean Absolute Error (MAE):** Measures the average magnitude of absolute errors, providing an intuitive interpretation in score points.
- **Root Mean Squared Error (RMSE):** Penalizes larger errors more heavily, highlighting significant prediction outliers.
- **R-squared (R^2):** Represents the proportion of variance in the dependent variable predictable from independent features.

Table 2: Model Performance Comparison Framework

Model	Val MAE	Val RMSE	Val R^2	Notes
Dummy Regressor	TBD	TBD	TBD	Baseline constant predictor
Ridge Regression	TBD	TBD	TBD	Linear baseline
Random Forest	TBD	TBD	TBD	Ensemble bagging
Extra Trees	TBD	TBD	TBD	Randomized trees
Gradient Boosting	TBD	TBD	TBD	Sequential boosting
XGBoost / CatBoost	TBD	TBD	TBD	Optimized boosting

16 Model Explainability and Feature Importance

To ensure transparency, BUNORA will utilize permutation importance to quantify the impact of individual features on model predictions. This approach identifies whether altitude, processing method, or country of origin exerts the strongest influence on predicted cup scores.

17 Proposed Frontend and Backend Architecture

The full-stack architecture separates concerns across three layers:

- **Frontend:** Built with React, TypeScript, and Tailwind CSS [5]. Provides a clean, responsive interface with form validation and prediction result visualization.
- **Backend:** Developed in Python using FastAPI [4]. Exposes RESTful endpoints (`/predict`, `/health`, `/model-info`) with Pydantic request validation and centralized error handling.
- **Persistence Layer:** Optional PostgreSQL database for logging prediction history and user interaction logs, entirely decoupled from the ML training artifacts.

18 Functional Requirements

- **FR1:** The system shall accept supported coffee metadata attributes through a web form.
- **FR2:** The backend shall validate all user-provided input against predefined schema constraints using Pydantic [4].
- **FR3:** The system shall process inputs through the serialized ML pipeline and return a predicted quality score.
- **FR4:** The frontend shall display the prediction result clearly alongside input summaries [5].
- **FR5:** The system shall handle missing or malformed inputs gracefully with informative error messages.

19 Non-Functional Requirements

- **NFR1 (Reproducibility):** Fixed random seeds and version-pinned dependencies shall ensure reproducible training and inference.
- **NFR2 (Performance):** API inference latency shall not exceed 500 milliseconds for standard prediction requests.
- **NFR3 (Usability):** The user interface shall be fully responsive across desktop and mobile viewports [5].
- **NFR4 (Maintainability):** Strict modular separation between frontend, backend, and machine learning components.

20 Technology Stack

Table 3: BUNORA Technology Stack

Component	Technologies
Machine Learning	Python, scikit-learn [3], pandas, NumPy, XGBoost, Joblib, MLflow
Backend API	Python, FastAPI [4], Uvicorn, Pydantic
Frontend	React [5], TypeScript, Tailwind CSS, Vite
Database	PostgreSQL (Optional history persistence)
DevOps & Tooling	Git, GitHub, Docker

21 System Architecture Diagram

The architecture is divided into two distinct operational paths: offline model training and online inference, complemented by an optional history persistence component.

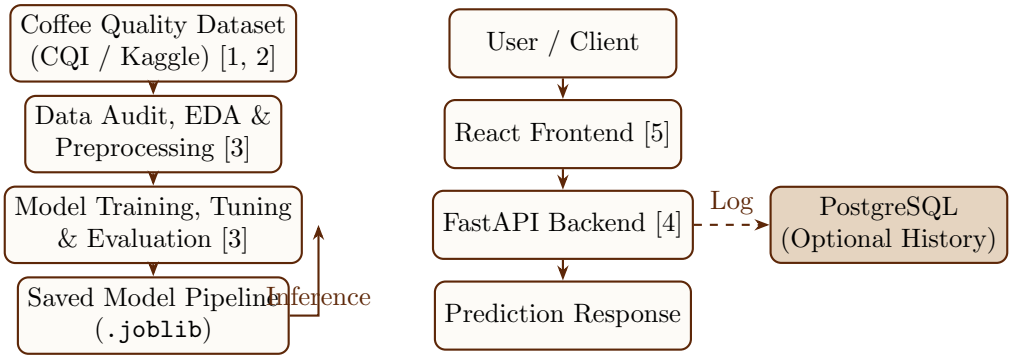


Figure 1: BUNORA Dual-Path System Architecture (Offline Training vs. Online Inference)

22 Expected Outputs and Deliverables

- Audited dataset and documented preprocessing scripts.
- Exploratory Data Analysis Jupyter notebooks and visualizations.
- Reproducible ML training, evaluation, and hyperparameter tuning code.
- Serialized best-performing ML pipeline.
- Fully functional FastAPI backend and React frontend.
- Comprehensive project report and final presentation.

23 Limitations, Risks, and Mitigation Strategies

- **Dataset Size and Imbalance:** The dataset is relatively small (∼1,340 samples) with heavy Arabica representation and few Robusta samples [1, 2]. *Mitigation:* Focus primarily on Arabica modeling and apply robust cross-validation.
- **Target Leakage Risks:** Unintentional inclusion of sensory sub-scores could artificially inflate accuracy. *Mitigation:* Strict feature auditing to exclude all post-tasting metrics from the primary feature set.
- **Generalization Limits:** Models trained on historical CQI data may not generalize perfectly to unrepresented coffee origins [1, 2]. *Mitigation:* Clearly communicate model boundaries and frame predictions as estimates rather than absolute guarantees.

24 Project Timeline

The project will be executed over an eight-week schedule as outlined below.

Table 4: Eight-Week Project Implementation Timeline

Week	Phase	Key Tasks
Week 1	Problem & Data	Problem definition, dataset acquisition, schema verification, data audit.
Week 2	EDA & Preprocessing	Exploratory Data Analysis, missing-value analysis, preprocessing pipeline.
Week 3	Baseline Models	Implementation of baseline models and initial machine learning algorithms.
Week 4	Model Tuning	Model comparison, cross-validation, and hyperparameter optimization.
Week 5	Evaluation	Final evaluation, error analysis, and feature importance explainability.
Week 6	Backend	FastAPI backend development, schema validation, and model integration.
Week 7	Frontend	React frontend development, API integration, and UI refinement.
Week 8	Finalization	Testing, documentation, packaging, final presentation preparation.

25 Conclusion

The BUNORA project presents a structured, academically rigorous approach to predicting coffee quality using machine learning. By combining a disciplined preprocessing pipeline, thorough algorithm comparison, and a modern full-stack web architecture, BUNORA demonstrates the practical application of AI in agricultural economics. Through careful management of target leakage and model limitations, the project delivers a reliable, interpretable tool that bridges data science and the global coffee industry.

References

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